

# Paper Title

Firstname Lastname and Firstname Lastname

Institute

**Abstract.** Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

**Keywords:** First keyword · Second keyword · Third keyword

## 1 Introduction

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy

pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et nunc am vulputate metus eu enim. Vestibulum pellentesque felis eu massa. TODO!

The remainder of the paper starts with a presentation of related work (Sect. 2). It is followed by a presentation of hints on L<sup>A</sup>T<sub>E</sub>X (Sect. 3). Finally, a conclusion is drawn and outlook on future work is made (Sect. 4).

## 2 Related Work

Winery [?] is a graphical modeling tool. The whole idea of TOSCA is explained by ? ].

## 3 LaTeX Hints

This section contains hints on writing LaTeX. It focuses on minimal examples, which can be directly adapted to the content

### 3.1 Handling of paragraphs

One sentence per line. This rule is important for the usage of version control systems. A new line is generated with a blank line. As you would do in Word: New paragraphs are generated by pressing enter. In LaTeX, this does not lead to a new paragraph as LaTeX joins subsequent lines. In case you want a new paragraph, just press enter twice! This leads to an empty line. In word, there is the functionality to press shift and enter. This leads to a hard line break. The text starts at the beginning of a new line. In LaTeX, you can do that by using two backslashes (\\).

This is rarely used.

Please do *not* use two backslashes for new paragraphs. For instance, this sentence belongs to the same paragraph, whereas the last one started a new one. A long motivation for that is provided at <http://loopspace.mathforge.org/HowDidIDoThat/TeX/VCS/#section.3>.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

638 One sentence per line.
639 This rule is important for the usage of version control systems.
640 A new line is generated with a blank line.
641 As you would do in Word:
642 New paragraphs are generated by pressing enter.
643 In LaTeX, this does not lead to a new paragraph as LaTeX joins
    subsequent lines.
644 In case you want a new paragraph, just press enter twice!
645 This leads to an empty line.
646 In word, there is the functionality to press shift and enter.
647 This leads to a hard line break.
648 The text starts at the beginning of a new line.
649 In LaTeX, you can do that by using two backslashes
    (\textbackslash\textbackslash).
650 \\
651 This is rarely used.
652
653 Please do \textit{not} use two backslashes for new paragraphs.
654 For instance, this sentence belongs to the same paragraph,
    whereas the last one started a new one.
655 A long motivation for that is provided at
    \url{http://loopspace.mathforge.org/HowDidIDoThat/TeX/VCS/#section.3}.

```

### 3.2 Notes separated from the text

The package mindflow enables writing down notes and annotations in a way so that they are separated from the main text.

---

This is a small note.

---

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

663 \begin{mindflow}
664 This is a small note.
665 \end{mindflow}

```

### 3.3 Handling TODOs

Markierter Text.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

671 \textmarker{Markierter Text.}

```

Bei `\textmarker` wird nur die Textfarbe geändert, da dies auch bei einigen Worten gut funktioniert.

Markierter Text.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```
677 \textcomment{Markierter Text.}{Kommentar dazu.}
```

Manuelle Markierung für Text, der seit der letzten Version geändert wurde.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```
681 \modified{Manuelle Markierung für Text, der seit der letzten
      Version geändert wurde.}
```

Das ist ein Text. Geänderter Text.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```
685 Das ist ein Text.
686 \change{FL1: Text angepasst}{Geänderter Text}.
```

Hier nur ein Kommentar.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```
690 Hier nur ein Kommentar\sidecomment{Kommentar}.
```

TODO!

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```
694 \todo{Hier muss noch kräftig Text produziert werden}
```

### 3.4 Hyphenation

L<sup>A</sup>T<sub>E</sub>X automatically hyphenates words. When using microtype, there should be fewer hyphenations than in other settings. It might be necessary to tweak the hyphenations nevertheless. Here are some hints:

In case you write “application-specific”, then the word will only be hyphenated at the dash. You can also write `applica\allowbreak{}tion-specific` (result: application-specific), but this is much more effort.

You can now write words containing hyphens which are hyphenated at other places in the word. For instance, `application=specific` gets application=specific. This is enabled by an additional configuration of the babel package.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

705 In case you write \enquote{application-specific}, then the word
      will only be hyphenated at the dash.
706 You can also write \verbiapplica\allowbreak{tion-specific}
      (result: applica\allowbreak{tion-specific}), but this is
      much more effort.
707
708 You can now write words containing hyphens which are hyphenated
      at other places in the word.
709 For instance, \verbiapplication="specific1 gets
      application="specific.
710 This is enabled by an additional configuration of the babel
      package.

```

### 3.5 Typesetting Units

Numbers can be written plain text (such as 100), by using the siunitx package as follows: 100  $\frac{\text{km}}{\text{h}}$ , or by using plain L<sup>A</sup>T<sub>E</sub>X (and math mode): 100  $\frac{\text{km}}{\text{h}}$ .

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

716 Numbers can be written plain text (such as 100), by using the
      \href{https://ctan.org/pkg/siunitx}{siunitx} package as
      follows:
717 \SI{100}{\km\per\hour},
718 or by using plain \LaTeX{} (and math mode):
719 $100 \frac{\mathit{km}}{h}$.

```

5 % of 10 kg

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

723 \SI{5}{\percent} of \SI{10}{kg}

```

Numbers are automatically grouped: 123 456.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

727 Numbers are automatically grouped: \num{123456}.

```

### 3.6 Surrounding Text by Quotes

Please use the “enquote command” to quote something. Quoting with “quote” or “quote” also works.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

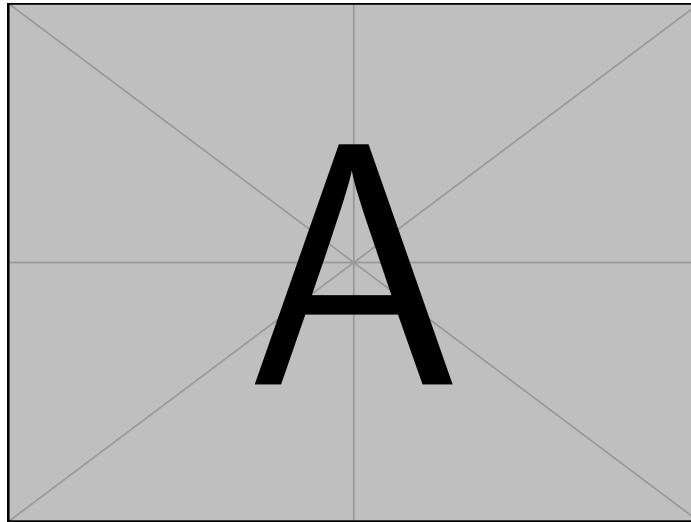
```

733 Please use the \enquote{enquote command} to quote something.
734 Quoting with "\enquote" or "\enquote" also works.

```

**3.7    Cleveref examples**

Cleveref demonstration: Cref at beginning of sentence, cref in all other cases.



**Fig. 1.** Example figure for cref demo

| Heading1 | Heading2 |
|----------|----------|
| One      | Two      |
| Thee     | Four     |

**Table 1.** Example table for cref demo

Figure 1 shows a simple fact, although Fig. 1 could also show something else.  
 Table 1 shows a simple fact, although Table 1 could also show something else.  
 Section 3.7 shows a simple fact, although Sect. 3.7 could also show something else.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```
765 \Cref{fig:ex:cref} shows a simple fact, although  
    \cref{fig:ex:cref} could also show something else.  
766  
767 \Cref{tab:ex:cref} shows a simple fact, although  
    \cref{tab:ex:cref} could also show something else.  
768  
769 \Cref{sec:ex:cref} shows a simple fact, although  
    \cref{sec:ex:cref} could also show something else.
```

### 3.8 Figures

Figure 2 shows something interesting.



**Fig. 2.** Simple Figure. Based on ? ].

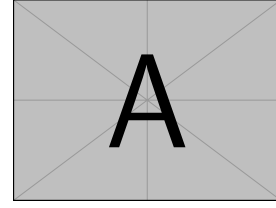
Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```
775 \Cref{fig:label} shows something interesting.  
776  
777 \begin{figure}  
778   \centering  
779   \includegraphics[width=.8\linewidth]{example-image-golden}  
780   \caption[Simple Figure]{  
781     Simple Figure.  
782     Based on \citet{mwe}.  
783   }  
784   \label{fig:label}  
785 \end{figure}
```

One can also have pictures floating inside text:



Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.



**Fig. 3.** A floating figure

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

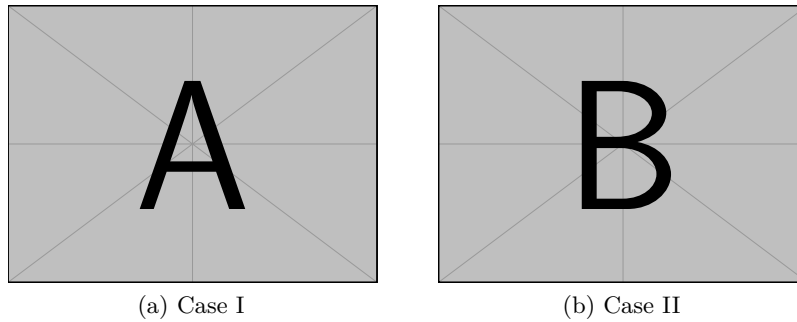
```

792 \begin{floatingfigure}{.33\linewidth}
793   \includegraphics[width=.29\linewidth]{example-image-a}
794   \caption{A floating figure}
795 \end{floatingfigure}
796 \lipsum[2]

```

### 3.9 Sub Figures

An example of two sub figures is shown in Fig. 4.



**Fig. 4.** Example figure with two sub figures.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

804 \begin{figure}[!b]
805   \centering
806   \subfloat[Case
      I]{\includegraphics[width=.4\linewidth]{example-image-a}%
807     \label{fig:first_case}}
808   \hfil
809   \subfloat[Case
      II]{\includegraphics[width=.4\linewidth]{example-image-b}%
810     \label{fig:second_case}}
811   \caption{Example figure with two sub figures.}
812   \label{fig:two_sub_figures}
813 \end{figure}

```

## 3.10 Tables

**Table 2.** Simple Table

| Heading1 | Heading2 |
|----------|----------|
| One      | Two      |
| Thee     | Four     |

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

819 \begin{table}
820   \caption{Simple Table}
821   \label{tab:simple}
822   \centering
823   \begin{tabular}{ll}
824     \toprule
825     Heading1 & Heading2 \\
826     \midrule
827     One      & Two      \\
828     Thee     & Four     \\
829     \bottomrule
830   \end{tabular}
831 \end{table}

```

**Table 3.** Table with diagonal line

|                       |                        |        |       |
|-----------------------|------------------------|--------|-------|
| Diag<br>Column Head I | Diag Column<br>Head II | Second | Third |
|                       |                        | foo    | bar   |

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```
835 % Source: https://tex.stackexchange.com/a/468994/9075
836 \begin{table}
837   \caption{Table with diagonal line}
838   \label{tab:diag}
839   \begin{center}
840     \begin{tabular}{|l|c|c|}
841       \hline
842       \diagbox[width=10em]{Diag \Column Head I}{Diag
843         Column\Head II} & Second & Third \\\hline
844       & foo & bar \\\hline
845     \end{tabular}
846   \end{center}
847 \end{table}
```

3.11 Source Code

Listing 1.1 shows source code written in XML. Line 2 contains a comment.

---

```
1 <listing name="example">
2   <!-- comment -->
3   <content>not interesting</content>
4 </listing>
```

---

**Listing 1.1.** Example XML Listing

---

```

1 <listing name="example">
2   Floating
3 </listing>

```

---

**Listing 1.2.** Example XML listing – placed as floating figure

---



---

```

1 {
2   key: "value"
3 }

```

---

**Listing 1.3.** Example JSON listing – placed as floating figure

---

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

855 \Cref{lst:XML} shows source code written in XML.
856 \Cref{line:comment} contains a comment.
857
858 \begin{lstlisting}[
859   language=XML,
860   caption={Example XML Listing},
861   label={lst:XML}]
862 <listing name="example">
863   <!-- comment --> (* \label{line:comment} *)
864   <content>not interesting</content>
865 </listing>
866 \end{lstlisting}

```

One can also add `float` as parameter to have the listing floating. Listing 1.2 shows the floating listing.

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

873 \begin{lstlisting}[
874   % one can adjust spacing here if required
875   % aboveskip=2.5\baselineskip,
876   % belowskip=-.8\baselineskip,
877   float,
878   language=XML,
879   caption={Example XML listing -- placed as floating figure},
880   label={lst:flXML}]
881 <listing name="example">
882   Floating
883 </listing>
884 \end{lstlisting}

```

One can also typeset JSON as shown in Listing 1.3.

---

```
1 public class Hello {
2     public static void main (String[] args) {
3         System.out.println("Hello World!");
4     }
5 }
```

---

**Listing 1.4.** Example Java listing

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```
890 \begin{lstlisting}[
891     float,
892     language=json,
893     caption={Example JSON listing -- placed as floating figure},
894     label={lst:json}]
895 {
896     key: "value"
897 }
898 \end{lstlisting}
```

Java is also possible as shown in Listing 1.4.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```
904 \begin{lstlisting}[
905     caption={Example Java listing},
906     label=lst:java,
907     language=Java,
908     float]
909 public class Hello {
910     public static void main (String[] args) {
911         System.out.println("Hello World!");
912     }
913 }
914 \end{lstlisting}
```

### 3.12 Itemization

One can list items as follows:

- Item One
- Item Two

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

922 \begin{itemize}
923   \item Item One
924   \item Item Two
925 \end{itemize}

```

One can enumerate items as follows:

1. Item One
2. Item Two

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

932 \begin{enumerate}
933   \item Item One
934   \item Item Two
935 \end{enumerate}

```

With paralist, one can even have all items typeset after each other and have them clean in the TeX document:

1. All these items... 2. ...appear in one line 3. This is enabled by the paralist package.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

942 \begin{inparaenum}
943   \item All these items...
944   \item ...appear in one line
945   \item This is enabled by the paralist package.
946 \end{inparaenum}

```

### 3.13 Other Features

The words “workflow” and “dwarflike” can be copied from the PDF and pasted to a text file.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

952 The words \enquote{workflow} and \enquote{dwarflike} can be
    copied from the PDF and pasted to a text file.

```

The symbol for powerset is now correct:  $\mathcal{P}$  and not a Weierstrass  $p$  ( $\wp$ ).  
 $\mathcal{P}(1, 2, 3)$

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

956 The symbol for powerset is now correct:  $\mathcal{P}$  and not a
      Weierstrass p ( $\wp$ ).
957
958  $\mathcal{P}(\{1,2,3\})$ 

```

Brackets work as designed: `<test>` One can also input backticks in verbatim text: ``test``.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of main.tex

```

962 Brackets work as designed:
963 <test>
964 One can also input backticks in verbatim text: \verb|`test`|.

```

## 4 Conclusion and Outlook

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

**Acknowledgments** Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which is placed just before the reference section in your document [? ].

All links were last followed on October 5, 2020.